

Apprentissage symbolique et statistique à l'ère du mariage pour tous

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Apprentissage : humain vs. machine

Les apprentissages d'un enfant

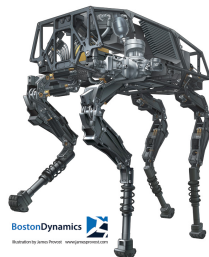
- marcher : un an
- parler : deux ans
- raisonner : le reste



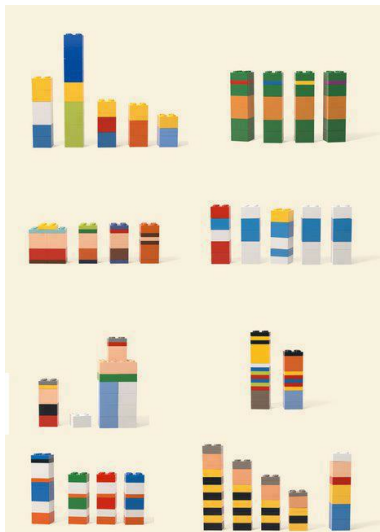
Apprentissage : humain vs. machine

Les apprentissages d'un enfant

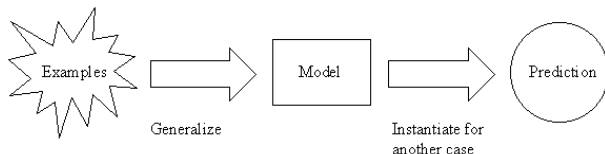
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Apprendre à raisonner



Symbolique et statistique : une opposition historique



Deux visions de l'apprentissage des machines

- Expliquer : trouver un model
- Prédire : améliorer des performances

Deux disciplines scientifiques concernées

- Informatique
- Statistique

Deux visions des statistiques

Table 1: An Outsider's Glossary: **NOT THE OPINION OF THE AUTHOR**

CONCEPT	BERKELEY	STANFORD
Statistics	Study of measurable functions	The bootstrap
Applied Statistics	Study of measurable functions in metric spaces	The bootstrap applied to MARS
Regression	Hilbert space projection	Special case of wavelet transform
A proof	Lucien LeCam says so	Charles Stein says so
Systems support	Spector, Licht, Kawin	A guy last seen somewhere in EE last month
Impediment to research progress	Data	Berkeley's research
One of our superstars	David Donoho	David Donoho
Most FAQ	Where is Donoho?	Where is Donoho?

Deux définitions de l'apprentissage

Arthur Samuel (1959)

Machine Learning : Field of study that gives computers the ability to learn without being explicitly programmed.



Tom Mitchell (The Discipline of Machine Learning, 2006)

“How can we build computer systems that automatically improve with experience, and what are the fundamental laws that govern all learning processes ?”

A computer program CP is said to learn from experience E with respect to some class of tasks T and performance measure P, if its **performance at tasks in T, as measured by P, improves** with experience E

Le problème d'apprentissage (CP, E, T, P)

- CP : le programme, c'est à dire l'algorithme qui apprend $\mathcal{A}$
- E : l'expérience, les données
 - ▶ les entrées x
 - ▶ les sorties : y
- P : la mesure de performance - le critère à optimiser
- T : la tâche

entrées :

apprentissage x_a
test x_t

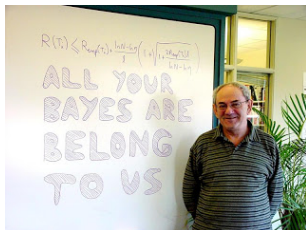
sorties :

apprentissage y_a
test y_t
renforcement r

tâche T	type	données E	critère P
Marketing	apprentissage non supervisé	x_a	information
OCR	apprentissage supervisé	x_a, y_a	cout 0/1
Recommandation	discrimination (<i>classification</i>)	$y_a \in \{0, 1\}$	mesure cumulative de
	ordonnancement (<i>ranking</i>)	$y_i < y_j$	qualité de tri (NDCG)
Prévision météo	régression	$y_i \in \mathbb{R}$	carrées des écarts
Génomique	apprentissage semi supervisé	(x_a, y_a, x_t)	
Jeux (Go)	apprentissage par renforcement	$x_{i,i=1,n}, r$	$\mathbb{E}(r)$

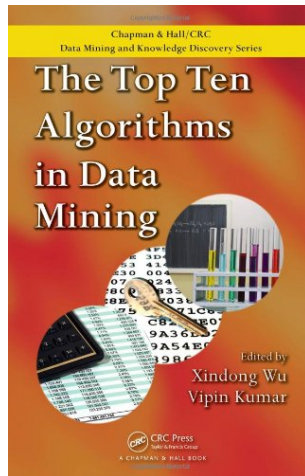
Les objectifs de l'apprentissage automatique

- Algorithmes d'apprentissage
 - ▶ bonne généralisation
 - ▶ généricité
 - ▶ passage à l'échelle
- Questions théoriques (Vapnik's Book, 1982 - Valiant, 1984)
 - ▶ apprenabilité, sous quelles conditions ?
 - ▶ complexité (en temps, en échantillon)



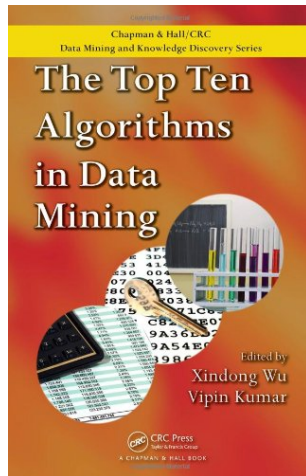
Top 10 algorithmes en fouille de données

- Arbres de décision (et les forêts)
 - ▶ C4.5,
 - ▶ CART,
- SVM,
- AdaBoost,
- kNN,
- Bayésien
 - ▶ k-Means,
 - ▶ Apriori,
 - ▶ EM,
 - ▶ Naive Bayes,
- PageRank,
-

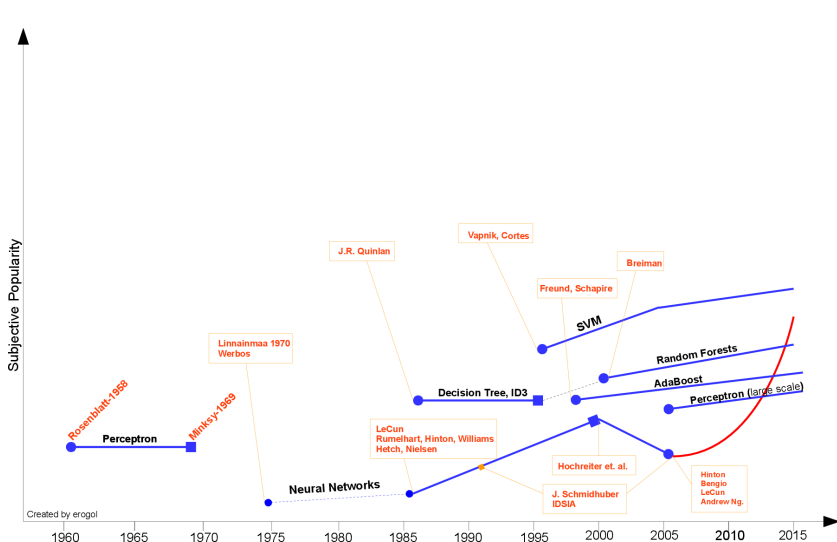


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- en 2014 un 11 ème : deep networks



Une brève histoire des modes en apprentissage



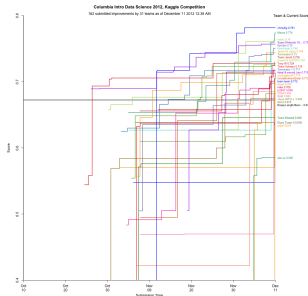
Faire de la science un sport !

- Netflix : un million de dollar pour deux ans de travail
- ...au service des entreprises : Datascience.net
 - ▶ SNCF : Prédiction de la fréquentation des gares SNCF en Ile-de-France
 - ▶ GDF Suez : Modéliser les consommations électriques de sites B2B
- Kaggle : *Go from Big Data to Big Analytics*
 - ▶ Higgs Boson Machine Learning Challenge proposé par le *Paris-Saclay Center for Data Science*



Leaderboard **10.05%** Display top 20 leaders.

Rank	Team Name	Best Score	% Improvement	Last Submit Time
1	BellKor's Pragmatic Chaos	0.8558	10.05	2009-06-26 18:42:37
Grand Prize - RMSE <= 0.8563				
2	PragmaticTheory	0.8582	9.80	2009-06-25 22:15:51
3	BellKor in BigChaos	0.8590	9.71	2009-05-13 08:14:09
4	Grand Prize Team	0.8593	9.68	2009-06-12 08:20:24
5	Dace	0.8604	9.56	2009-04-22 05:57:03
6	BigChaos	0.8613	9.47	2009-06-23 23:06:52



Retour aux applications : deep learning & big data

- Quelques challenges + la traduction automatique : *deep learning strikes back*
- web : google, facebook...
- Marketing : Walmart & big data
 - ▶ volume
 - ▶ variété
 - ▶ vitesse



New trends in ML fashion : big data

ICML 2014 workshops

- Designing Machine Learning Platforms for **Big Data**
Xiangxiang Meng, Wayne Thompson, Xiaodong Lin
- New Learning Frameworks and Models for **Big Data**
Massih-Reza Amini, Eric Gaussier, James Kwok, Yiming Yang
- Deep Learning Models for Emerging **Big Data** Applications
Shan Suthaharan, Jinzhu Jia
- Unsupervised Learning for Bioacoustic **Big Data**
Hervé Glotin, P. Dugan, F. Chamroukhi, C. Clark, Yann LeCun
- Knowledge-Powered **Deep Learning for Text Mining**
Bin Gao, Scott Yih, Richard Socher, Jiang Bian
- Optimizing Customer Lifetime Value in **Online Marketing**
Georgios Theodorou, Mohammad Ghavamzadeh, Shie Mannor

Comment traiter ces *big data* : data science

this joke makes perfect sense

4 January 2013 12:28

Big data is like teenage sex: everyone talks about it, nobody really knows how to do it, everyone thinks everyone else is doing it, so everyone claims they are doing it



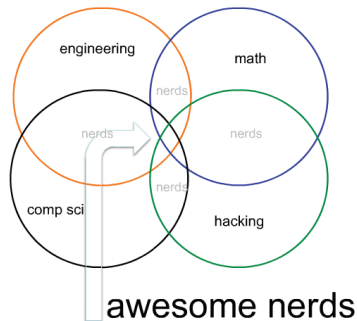
@博海ishi

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New trends in ML fashion : data science



Data science?



Data Scientist Voted Sexiest Job of 21st Century

The Rise of Data Scientists

BEFORE



nobody cared for a
"math geek" in parties.

NOW



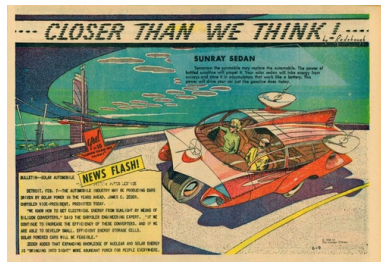
People love ~~math~~geeks!
data scientists!
RK

New trends in ML fashion : big data hiring boom



Quelques prédictions... à propos du futur

- Data science - Applications
 - ▶ multi compétences
 - ▶ chaine de traitements
- Outils pour l'apprentissage
 - ▶ l'apprentissage sans paramètres (off-the-shelf)
 - ▶ passage à l'échelle (3v - big data)
- Algorithmes d'apprentissage
 - ▶ optimisation (big data, non convexe)
 - ▶ dynamique (interactions)
 - ▶ apprendre à apprendre (transfert)
- Théorie de l'apprentissage
 - ▶ la nature de l'information



Références

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